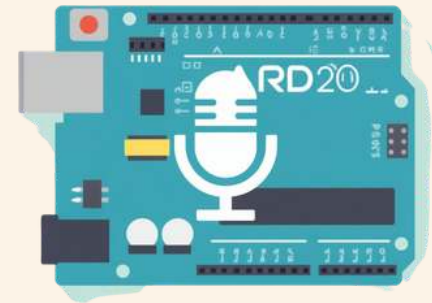
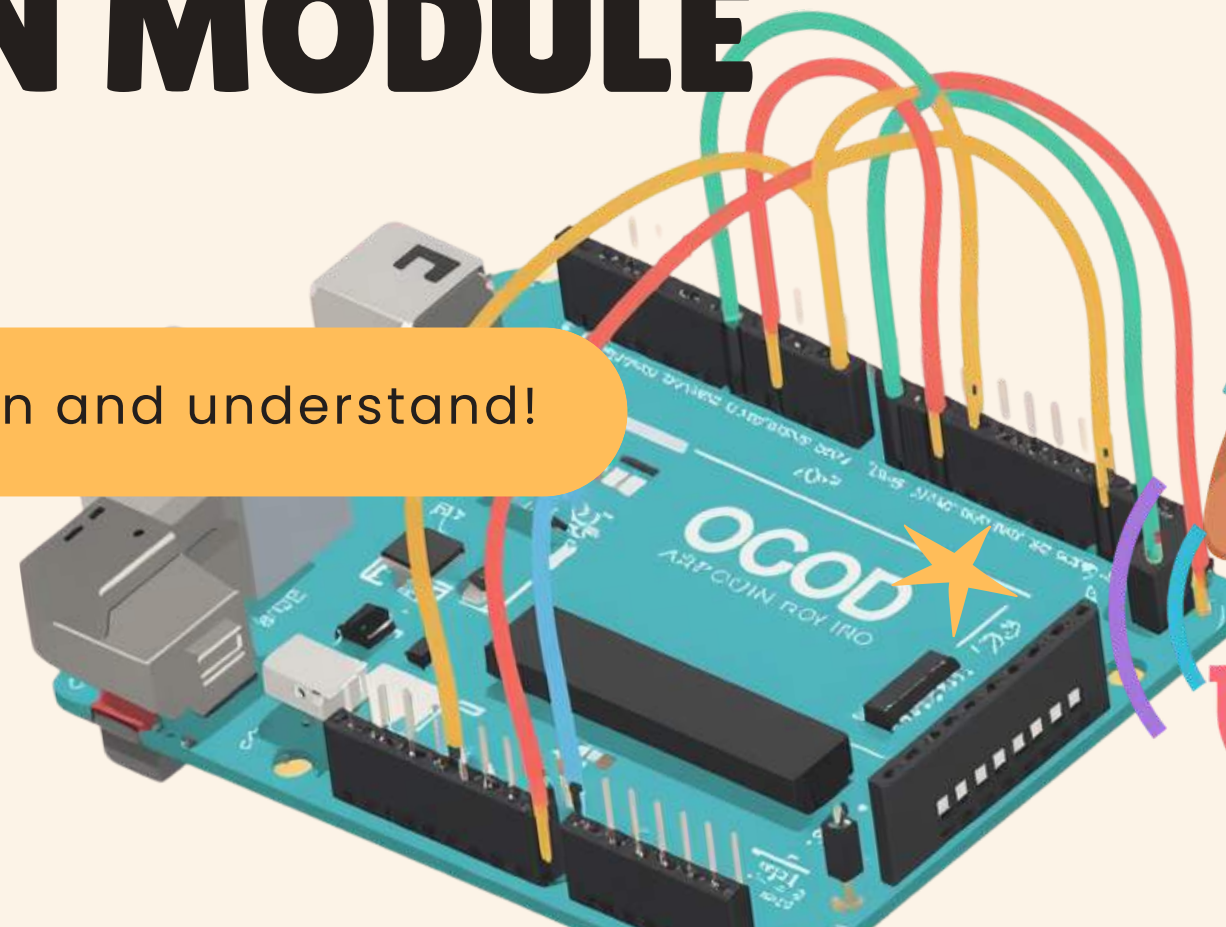




DISCOVERING **GRADE 6 ARDUINO – VOICE RECOGNITION MODULE**

Learn how your Arduino can listen and understand!



LET'S EXPLORE VOICE RECOGNITION!

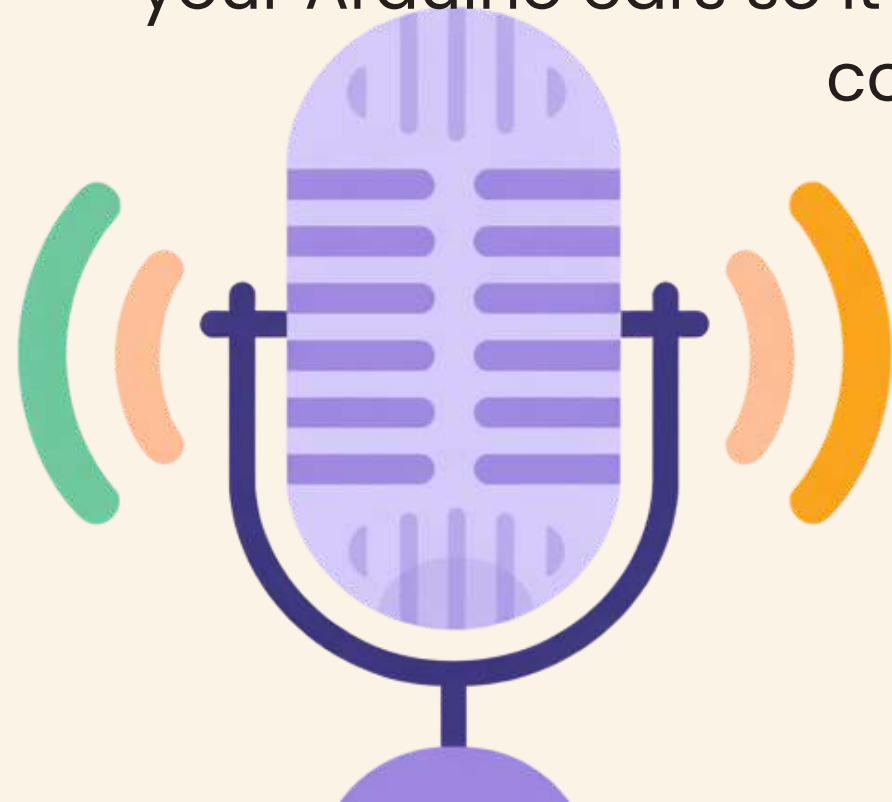


Today, we will discover how computers can understand our voices!

Get ready for an exciting journey into the world of voice technology and Arduino!

WHAT IS A VOICE RECOGNITION MODULE?

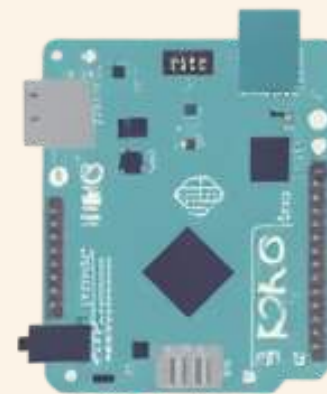
A voice recognition module is a small device that listens to your voice and tells the Arduino what you said. It's like giving your Arduino ears so it can hear and understand your commands!



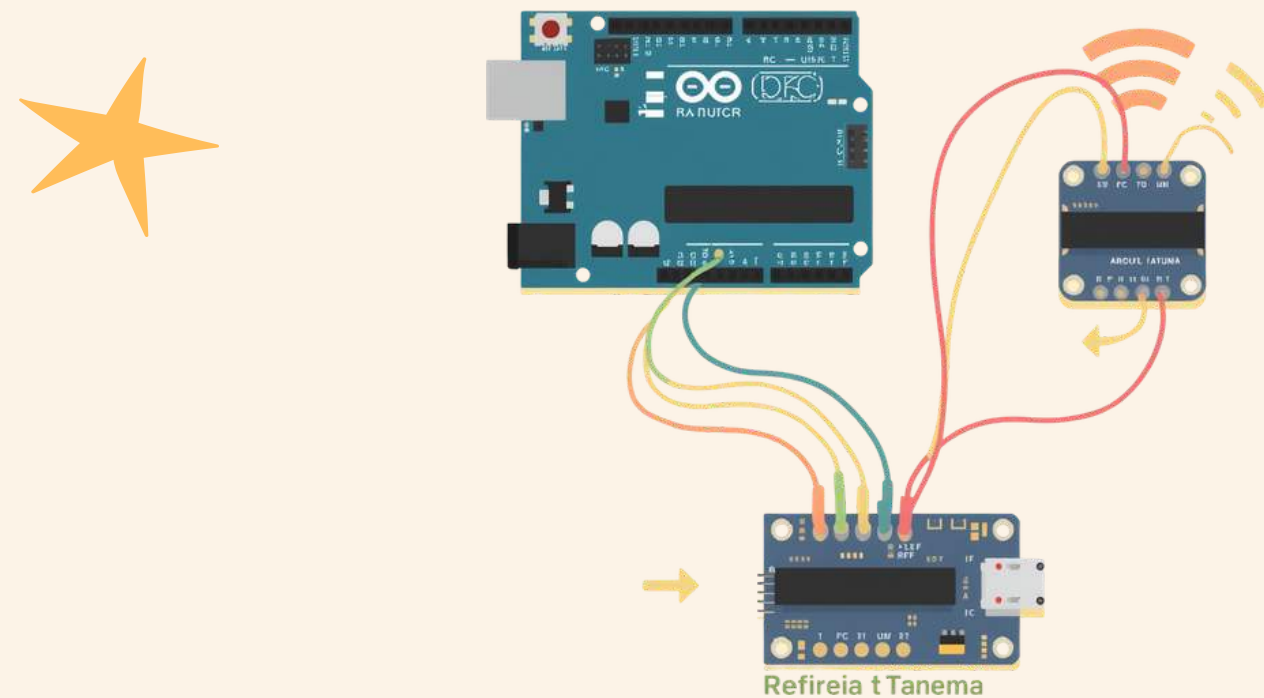
HOW DOES VOICE RECOGNITION WORK?

Voice Recognition

The module hears your voice, changes it to digital signals, and matches it to commands it knows. It's like magic — you speak, and your Arduino understands!



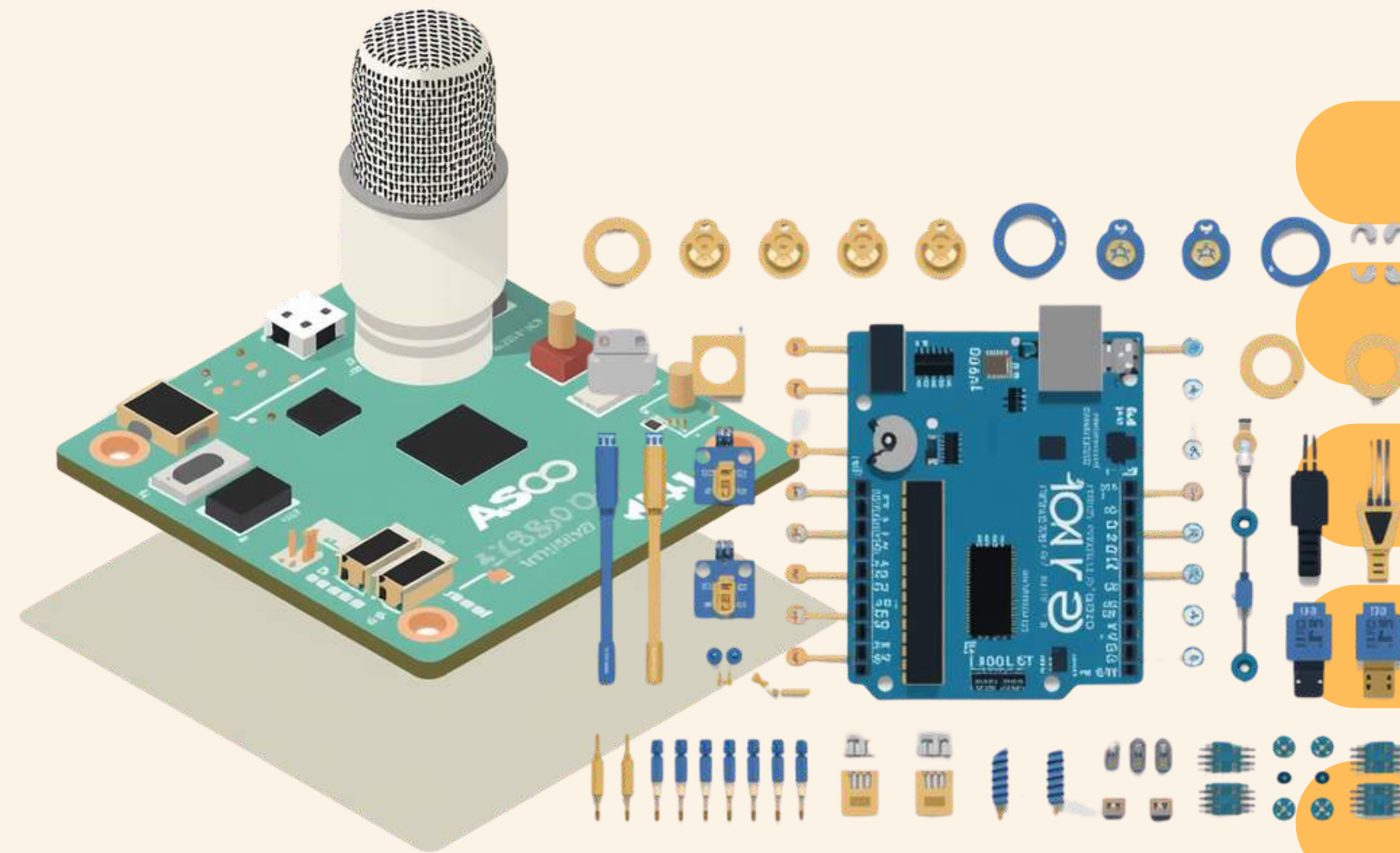
ARDUINO AND VOICE MODULE TOGETHER



The Arduino uses the voice recognition module to follow your voice commands and do cool things!

When you speak, the voice module listens and sends signals to the Arduino, which then performs actions like turning on lights or moving robots!

Components You Need



Voice recognition module

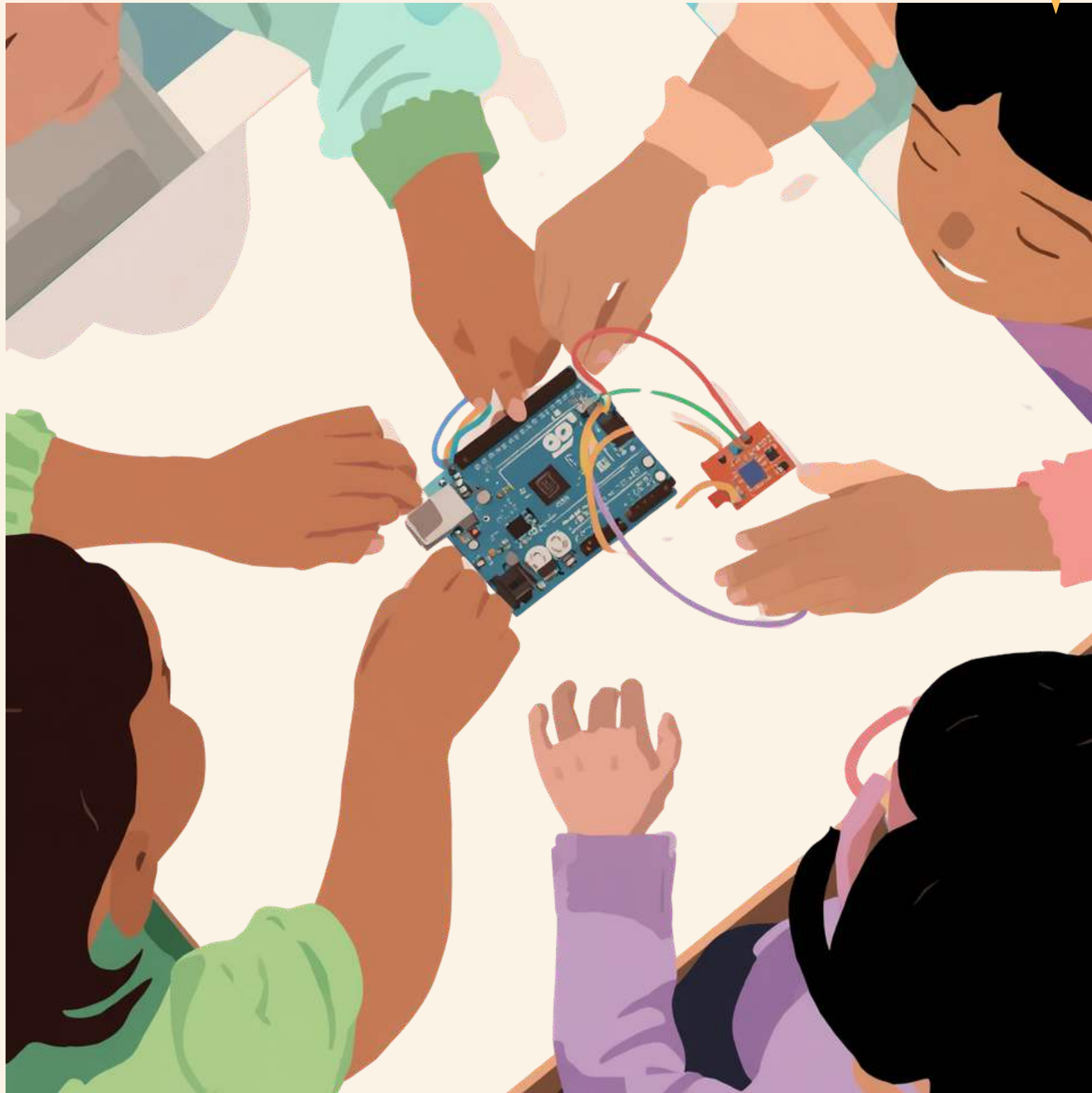
Arduino board (Uno or similar)

Connecting wires (jumper wires)

Power supply (USB or battery)

Speaker (optional for sound output)

Gather these items before starting your voice recognition project. Ask your teacher if you need help finding any parts!



Basic Setup and Connections

Connect voice module to Arduino pins (power, ground, TX, RX)

Power the board using USB or battery

Optional: Connect speaker for sound output

Double-check all connections before turning on

Ask your teacher if you need help!

SIMPLE CODING EXAMPLE

Here's a simple Arduino code to recognize a voice command and turn on an LED.

```
void setup() {  
  Serial.begin(9600);  
  pinMode(13, OUTPUT);  
}  
  
void loop() {  
  if (voiceCommand == "LED ON") {  
    digitalWrite(13, HIGH);  
  }  
  if (voiceCommand == "LED OFF") {  
    digitalWrite(13, LOW);  
  }  
}
```

Don't worry, your teacher will help explain this!

FUN PROJECT IDEAS



**Let's see some fun projects you
can build with voice recognition!**

Get ready to use your imagination and create amazing voice-controlled inventions with Arduino!



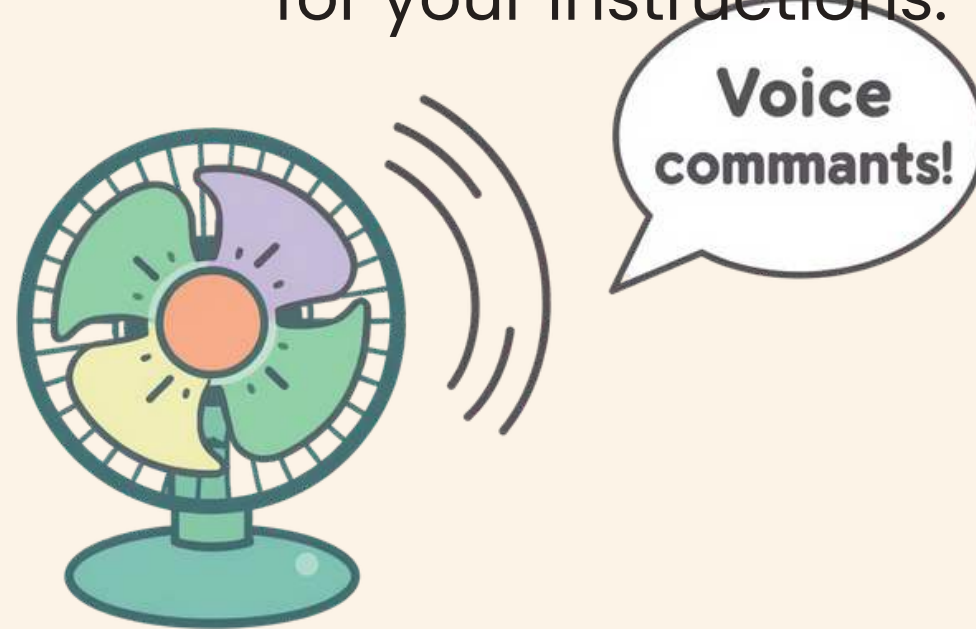
PROJECT IDEA 1: VOICE-CONTROLLED LIGHT

Control your room light just by saying "Turn on the light!" This simple project helps you practice using voice commands with your Arduino. You'll learn how to connect the voice module to a light and make it respond to what you say. It's a great first step to becoming a voice-control expert!



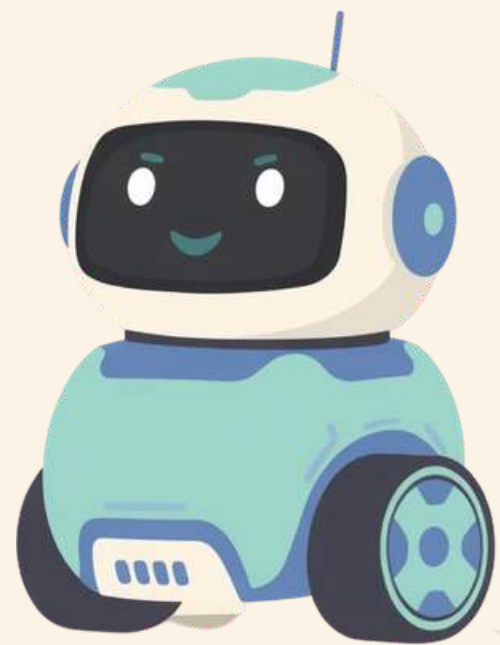
PROJECT IDEA 2: VOICE-ACTIVATED FAN

Say "Fan on" or "Fan off" to control a small fan with your voice! This fun project helps you learn about motors and how voice commands can make things move. You will connect a small fan to your Arduino and program it to listen for your instructions.



PROJECT IDEA 3: VOICE-ENABLED ROBOT

Make a small robot move forward or backward using just your voice! Say "Go forward" or "Go back" to control your robot. This project combines everything you've learned about coding and electronics into one exciting creation!



Safety and Best Practices

Always ask an adult before using electronics

Double-check connections before powering on

Keep your workspace clean and dry

Don't overload the Arduino with too many parts

Handle all components gently and carefully



Real-World Applications

Smart home assistants (Alexa, Google Home)

Voice-controlled robots in factories

Hands-free devices for people with disabilities

Voice commands in cars for safety



SUMMARY — YOU CAN DO IT!



Now you know how voice recognition works with Arduino! Try these projects and become a young inventor!

You learned about voice modules, connections, coding, and fun projects.
Keep exploring and creating amazing things with technology!



WE LOVED LEARNING WITH YOU!

**THANK YOU AND
QUESTIONS?**

Thank you for listening! Any questions?

